

Rhino Bank Infrastructure Memorandum

Tees Valley Metro

A Regional Rapid Rail Network Centred on Darlington

Prepared for: Rhino Bank Infrastructure Division

Executive Summary

The Tees Valley already possesses one of Britain's greatest railway assets:

- the original Stockton & Darlington Railway
- an extensive heavy rail network
- multiple freight corridors
- an electrified East Coast Main Line
- major stations at Darlington, Middlesbrough, Stockton, Hartlepool and Redcar.

However, these lines largely operate as conventional regional rail with relatively low frequencies. Previous regional strategies have identified Darlington as the key interchange for faster and more frequent local services across Tees Valley, County Durham and North Yorkshire.

Rhino Bank proposes converting this network into a "**Tees Valley Metro**": a high-frequency, S-Bahn/RER-style system using upgraded existing railway infrastructure with selective new alignments, tunnels and bypasses to increase speed, reliability and capacity while avoiding the cost of building an entirely new metro.

Vision

Darlington becomes the regional "Grand Central" of Tees Valley.

Every town within approximately 30 miles would receive a train every **10–15 minutes**, all timed to connect at Darlington with LNER, CrossCountry and TransPennine services.

The objective is for rail to become the default transport mode for commuting, education and leisure.

Proposed Metro Network

Red Line

Darlington → Eaglescliffe → Stockton → Middlesbrough → Redcar → Saltburn

Length:

66 km

Frequency:

Every 10 minutes

Blue Line

Darlington

↓

North Road

↓

Shildon

↓

Bishop Auckland

↓

Future extension to Barnard Castle

Frequency

Every 15 minutes

Green Line

Darlington

↓

Northallerton

↓

Yarm

↓

Eaglescliffe

↓

Darlington

Circular service

Every 15 minutes

Orange Line

Darlington



Hartlepool



Seaton Carew



Middlesbrough



Darlington

Every 15 minutes

Airport Express

Darlington



Teesside Airport



Stockton



Middlesbrough

Every 20 minutes

Infrastructure Upgrades

1. Darlington Metro Hub

Existing redevelopment has delivered additional platforms, a transport interchange and improved capacity. Rhino Bank proposes extending this into a true metro hub with dedicated suburban platforms and through-running services.

Investment:

£750 million

Including:

- dedicated metro platforms

- underground passenger concourse
 - automatic train operation capability
 - integrated bus station
 - bicycle hub
 - retail development
-

2. Darlington–Stockton Fast Line

Current alignment follows Victorian geometry.

Proposal:

- straighten several curves
- new bypass around slower sections
- remove level crossings
- modern signalling

Length improved:

15 km

Cost:

£900 million

Maximum speed:

Current

60–75 mph

Future

100 mph

Journey

Darlington → Stockton

Current

17 minutes

Future

10 minutes

Time saving:

7 minutes

3. Darlington–Middlesbrough Express Alignment

Construct approximately 12 km of new railway bypassing the slowest sections approaching Middlesbrough.

This includes:

- one tunnel
- two viaducts
- new flying junction

Cost

£1.4 billion

Journey

Current

34 minutes

Future

20 minutes

Saving

14 minutes

4. Teesside Urban Tunnel

New underground section beneath central Middlesbrough.

Approximately

4 km

Stations

- University
- Town Centre
- Riverside
- Docklands

Cost

£1.8 billion

Benefits

- removes conflicting junctions
 - separates freight
 - allows metro frequencies
 - unlocks redevelopment land
-

5. Redcar Coastal Upgrade

Improve speeds by:

- straightening curves
- replacing junctions
- removing speed restrictions

Investment

£600 million

Journey

Current

23 minutes

Future

15 minutes

6. Hartlepool Bypass

Construct approximately 8 km inland bypass.

Purpose:

- faster metro
- freight segregation
- more local stopping services

Investment

£950 million

Rolling Stock

Fleet

60 electric metro trains

Features

- walk-through carriages
- 110 mph capability
- regenerative braking
- battery operation for unelectrified sections
- level boarding

Investment

£720 million

Service Frequency

Route	Current	Metro
Darlington–Stockton	2 tph	6 tph
Darlington–Middlesbrough	2 tph	6 tph
Darlington–Hartlepool	1 tph	4 tph
Darlington–Redcar	2 tph	6 tph
Darlington–Saltburn	1 tph	4 tph
Darlington–Bishop Auckland	2 tph	4 tph

Journey Time Improvements

Journey	Today	Metro	Saving
Darlington–Stockton	17 min	10 min	7 min
Darlington–Middlesbrough	34 min	20 min	14 min
Darlington–Hartlepool	38 min	24 min	14 min
Darlington–Redcar	47 min	31 min	16 min
Darlington–Saltburn	56 min	39 min	17 min
Darlington–Bishop Auckland	23 min	15 min	8 min

Passenger Capacity

Current annual rail journeys

Approximately

8–10 million

Projected Metro

24–28 million

Approximately three times current demand.

Economic Impact

The Tees Valley Combined Authority has previously estimated that rail investment centred on Darlington could generate around **£130 million per year** in economic benefits and support thousands of jobs.

Rhino Bank's more ambitious metro proposal could support substantially larger gains through wider labour-market access and higher service frequencies. Illustrative long-term impacts include:

Measure	Estimated Impact
Construction Jobs	32,000
Permanent Jobs	9,500
Additional Annual GVA	£1.8–2.3 billion
New Businesses	2,600
New Housing	24,000 homes
Annual Visitors	+3.5 million
These figures are strategic planning estimates rather than forecasts.	

Capital Cost

Investment	Cost
Darlington Hub	£0.75bn
Fast Line	£0.90bn
Middlesbrough Bypass	£1.40bn
Urban Tunnel	£1.80bn
Redcar Upgrade	£0.60bn
Hartlepool Bypass	£0.95bn
Rolling Stock	£0.72bn
Digital Signalling & Electrification	£1.10bn

Total Programme

£8.22 Billion

Long-Term Vision

The Tees Valley Metro would be a regional rapid-transit network that builds on existing heavy rail infrastructure rather than replacing it. By combining targeted new tunnels, bypasses, straighter alignments and high-frequency operations, the network would place Darlington at the centre of a metropolitan rail system linking Teesside, County Durham and North Yorkshire.

The proposal complements recent investment in Darlington Station and aligns with longstanding regional ambitions to improve local rail connectivity, while extending them into a larger integrated transport vision designed to increase productivity, attract investment and support long-term economic growth across the Tees Valley.